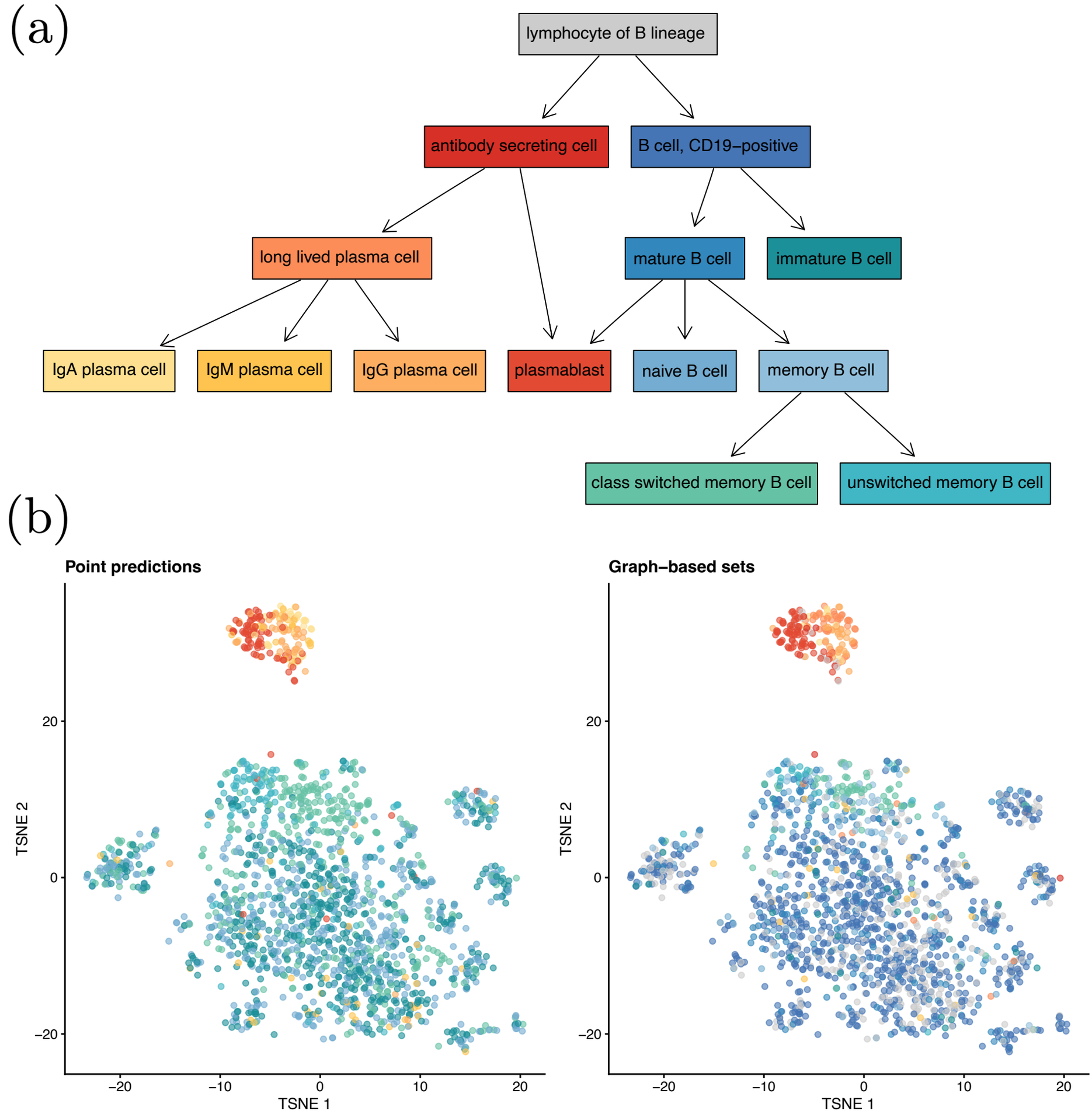
(a)
lymphocyte of B lineage
antibody secreting cell
B cell, CD19−positive
long lived plasma cell
mature B cell
immature B cell
IgA plasma cell
IgM plasma cell
IgG plasma cell
plasmablast
naive B cell
memory B cell
class switched memory B cell
unswitched memory B cell
(b)
Point predictions
Graph−based sets
TSNE 2
TSNE 1
20
0
−20
−20
−10
0
10
20